

INFLUENCE OF DIFFERENT RESISTANCE EXERCISE LOADING SCHEMES ON MECHANICAL POWER OUTPUT IN WORK TO REST RATIO – EQUATED AND – NONEQUATED CONDITIONS

CAETANO A. PAULO,^{1,2,3} HAMILTON ROSCHEL,¹ CARLOS UGRINOWITSCH,¹ RONALDO KOBAL,¹ AND VALMOR TRICOLI¹

¹Laboratory of Neuromuscular Adaptations to Strength Training, School of Physical Education and Sport, University of São Paulo, São Paulo, Brazil; ²Department of Physical Education, University Nove de Julho, São Paulo, Brazil; and ³Department of Physical Education, University Paulista–UNIP, São Paulo, Brazil

ABSTRACT

Paulo CA, Roschel H, Ugrinowitsch C, Kobal R and Tricoli V. Influence of different resistance exercise loading schemes on mechanical power output in work to rest ratio–equated and –nonequated conditions. *J Strength Cond Res* 26(5): 1308–1312, 2012—It is well known that most sports are characterized by the performance of intermittent high-intensity actions, requiring high muscle power production within different intervals. In fact, the manipulation of the exercise to rest ratio in muscle power training programs may constitute an interesting strategy when considering the specific performance demand of a given sport modality. Thus, the aim of this study was to evaluate the influence of different schemes of rest intervals and number of repetitions per set on muscle power production in the squat exercise between exercise to rest ratio–equated and –nonequated conditions. Nineteen young males (age: 25.7 ± 4.4 years; weight: 81.3 ± 13.7 kg; height: 178.1 ± 5.5 cm) were randomly submitted to 3 different resistance exercise loading schemes, as follows: short-set short-interval condition (SSSI; 12 sets of 3 repetitions with a 27.3-second interval between sets); short-set long-interval condition (SSLI; 12 sets of 3 repetitions with a 60-second interval between sets); long-set long-interval (LSLI; 6 sets of 6 repetitions with a 60-second rest interval between sets). The main finding of the present study is that the lower exercise to rest ratio protocol (SSLI) resulted in greater average power production (601.88 ± 142.48 W) when compared with both SSSI and LSLI (581.86 ± 113.18 W; 578 ± 138.78 W, respectively). Additionally, both the exercise to rest ratio–equated conditions

presented similar performance and metabolic results. In summary, these findings suggest that shorter rest intervals may fully restore the individual's ability to produce muscle power if a smaller exercise volume per set is performed and that lower exercise to rest ratio protocols result in greater average power production when compared with higher ratio ones.

KEY WORDS rest interval, muscle power, effort to pause ratio, training density

INTRODUCTION

Training programs designed to increase muscle power commonly require maximal rates of force development (i.e., $\Delta F/\Delta t$) during multiple sets of repeated maximal-effort movements (8,15,18). Literature shows that a usual muscle power training program comprises the use of (a) 3–6 repetitions per set; (b) an overload ranging from 30 to 60% 1 repetition maximum (1RM); (c) high-velocity exercise movement; and d) 3- to 5-minute rest intervals (5,6,10,13).

Despite the previous recommendations, information regarding the manipulation of the exercise to rest ratio in muscle power training programs is still scarce. Manipulating this variable might constitute an interesting strategy when considering the specific performance demand of a given sport modality and, as a consequence, the appropriateness of the training stimulus (i.e., maintenance of the power production). In fact, it is well known that most sports are characterized by the performance of intermittent high-intensity actions, thus requiring high muscle power production within different intervals. Therefore, exercise to rest ratio adjustment is an important tool in training design.

In this regard, Abdessemed et al. (1) compared the muscle power output in the bench press exercise during 3 different recovery conditions using a fixed exercise load and constant exercise volume. The authors demonstrated that the shorter rest interval (i.e., 60-second pauses between sets) resulted in

Address correspondence to Hamilton Roschel, hars@usp.br.
26(5)/1308–1312

Journal of Strength and Conditioning Research
© 2012 National Strength and Conditioning Association

decreased power output from sets 4 to 10 when compared with longer rest interval protocols (i.e., 180 or 300 seconds between sets). Even though these data are in accordance with other reports (13,16), it is important to note the evident differences in the exercise to rest ratio between the experimental conditions. Namely, the 60-second rest interval protocol resulted in a 3- to 5-fold greater exercise to rest ratio when compared with protocols comprising longer intervals (i.e., 180- and 300-second rest interval protocols, respectively), which may have influenced the results. Hence, it seems plausible to suggest that in an exercise to rest ratio-equated and training load-equated condition, the manipulation of the rest interval may play a role in the muscle power output maintenance.

Supporting this notion, Lawton et al. (9) demonstrated that the adoption of different rest interval schemes (in protocols that were equated for the exercise to rest ratio) allowed for a greater power output when compared with a traditional high-intensity, continuous-repetition weight training protocol of equal volume and intensity but of different exercise to rest ratio. However, there is a paucity of data on the effects of the manipulation of the exercise loading scheme in exercise to rest ratio-equated and -nonequated conditions and its effects in maintaining the muscle power output.

Furthermore, it is important to emphasize that in both Abdessemed et al. (1) and Lawton et al. (9) studies, the resistance exercises protocol did not mimic a typical power training session in which exercises had to be performed as fast as possible while maintaining proper form and using training loads within the maximum power zone (i.e., 30–60% 1RM load). Thus, to the best of our knowledge, there is no report in the literature regarding the exercise to rest ratio in resistance exercise sessions devoted to develop muscle power. Maintaining a fairly constant muscle power output throughout a training session seems to be critical to optimize the training volume and hence the improvements in functional performance.

In the light of these, the aim of this study was to evaluate the influence of different schemes of rest intervals and number of repetitions per set on muscle power production between exercise to rest ratio-equated and -nonequated conditions.

METHODS

Experimental Approach to the Problem

To evaluate the influence of different schemes of rest intervals and number of repetitions per set on muscle power production between equated and nonequated exercise to rest ratio conditions, the subjects were randomly submitted to 3 different experimental sessions on separate occasions (72–96 hours apart). Before the experimental sessions, 3 familiarization sessions (to the squat maximum dynamic strength test [1RM] and experimental conditions) took place. The experimental conditions were equated for total exercise volume and intensity (i.e., training load). To evaluate the effect of the rest interval in nonequated exercise to rest ratio conditions, we compare

a short-set and long-interval (SSLI) training scheme to a short-set and short-interval (SSSI) scheme. It is important to note that what we refer to as long or short rest interval should be confined within the context of the present study. Long rest intervals are usually understood as a period of time between 5 and 7 minutes that allows full restoration of the ATP stores, which is not the case in our study. We also evaluated the effect of the number of repetitions per set in nonequated exercise to rest ratio conditions (SSLI vs. long-set long-interval [LSLI] schemes). Additionally, we were also interested in the effects of different schemes of rest intervals and number of repetitions per set between exercise to rest ratio-equated conditions (SSSI vs. LSLI schemes). The average power output and blood lactate concentration were assessed throughout each experimental condition.

Subjects

Nineteen young men (age: 25.7 ± 4.4 years; weight: 81.3 ± 13.7 kg; height: 178.1 ± 5.5 cm) took part in the study. Subjects were not enrolled in any strength training program or taking any dietary supplement for at least 6 months before the study. None of them presented any musculoskeletal or cardiovascular disorders that precluded their participation in the study. The study was approved by the local ethical committee, and informed consents were obtained from all the subjects.

Familiarization

All the subjects completed 3 familiarization sessions. During the familiarization sessions, subjects performed a general warm-up consisting of 5 minutes running at $9 \text{ km} \cdot \text{h}^{-1}$ on a treadmill followed by 3 minutes of whole-body light stretching exercises. After warming up, the subjects were familiarized with the squat 1RM test protocol and to the 3 different experimental conditions (consisting of power training sessions with different rest intervals and number of repetitions per set performed at the highest velocity possible). The interday variance was $<5\%$ between familiarization sessions 2 and 3. Body position and foot placement were determined with measuring tapes fixed on the bar and on the ground, respectively. In addition, a wooden seat with adjustable heights was placed behind the subject to keep the bar displacement and knee angle ($\sim 90^\circ$) constant on each squat repetition. Subjects' positioning were recorded during the familiarization sessions and reproduced throughout the study.

Maximum Dynamic Strength Test (One Repetition Maximum)

Two days after the last familiarization session, 1RM strength for the lower limbs was assessed using the squat exercise on a conventional Smith machine (Cybex, Medway, MA, USA). Testing protocol followed the suggestions proposed by Brown and Weir (4). In brief, subjects ran for 5 minutes on a treadmill at $9 \text{ km} \cdot \text{h}^{-1}$ followed by lower limb stretching exercises and 2 squat warm-up sets. During the first set, subjects performed 5 repetitions with 50% of the estimated 1RM. In the second set, they performed 3 repetitions with 70% of the estimated 1RM, with 3-minute intervals between

TABLE 1. Experimental conditions.*

Experimental condition	No. of sets $\times$ repetitions (60% 1RM)	Rest interval between sets (seconds)	Total no. of repetitions	Total rest time (seconds)	Exercise to rest ratio
LSLI	6 $\times$ 6	60	36	300	36/300 = 0.12
SSSI	12 $\times$ 3	27.3	36	300	36/300 = 0.12
SSLI	12 $\times$ 3	60	36	660	36/660 = 0.05

*LSLI = long-set long-interval condition; SSSI = short-set short-interval condition; SSLI = short-set long-interval condition.

them. After the second warm-up set, subjects rested for 3 minutes. Then, they had up to 5 trials to achieve the 1RM load (i.e., maximum weight that could be lifted once with the proper technique), with a 3-minute interval between trials.

Experimental Protocols

All the subjects underwent 3 different experimental conditions in a random fashion. The experimental conditions comprised the performance of maximum-velocity squat exercise. The exercise volume and intensity were equated between conditions. A total of 36 repetitions were performed at 60% 1RM with different schemes of rest intervals and number of repetitions per set. The SSSI comprised 12 sets of 3 repetitions with a 27.3-second interval between sets. The SSLI consisted of 12 sets of 3 repetitions with a 60-second interval between sets. These 2 conditions allowed for the comparison of different interval schemes on nonequated exercise to rest ratio conditions on muscle power production. The remaining experimental session consisted of 6 sets of 6 repetitions with a 60-second rest interval between sets (LSLI). The purpose of the comparison between SSLI and LSLI was to evaluate the effect of a different number of repetitions per set in nonequated exercise to rest ratio conditions. Further, the comparison between SSSI and LSLI allowed further insight regarding the manipulation of the rest interval and the number of repetitions per set on exercise to rest ratio-equated conditions. Table 1 illustrates the different experimental conditions. All the experimental sessions were conducted during the same time of the day (i.e., around 10 AM, 2 hours after breakfast on the average). Subjects were instructed to refrain from exercising 72 hours before the experimental sessions and to maintain their regular diet throughout the study.

Muscle Power Output

The average power produced during each experimental condition was assessed by a linear encoder (Peak Power; Cefise, São Paulo, Brazil). The equipment was attached to the Smith machine bar to register its position throughout the repetitions at a frequency of 50 Hz. A finite differentiation technique was used to estimate bar velocity and acceleration

(variability coefficient <3%). Then, force and power were calculated using standard procedures (3).

Blood Lactate Concentration

Blood samples (25 μ L) were obtained from the earlobe before and immediately after the completion of each experimental condition. Blood lactate concentration was measured by an electrochemical technique (Lactate Analyzer, Yellow Springs Instruments 2300 Stat Plus; Yellow Springs Instruments, Yellow Springs, OH, USA) after stabilization in sodium fluoride (5 mM).

Statistical Analyses

Results are presented as average $\pm$ SD. Data normality was assessed through Shapiro-Wilk test and standard visual inspection; all the variables presented normal distribution. The variance homogeneity was tested with the Levene's test. A Proc Mixed Model (SAS) was performed for average power and blood lactate concentration. Whenever a significant *F* value was obtained, a post hoc test with a Tukey's adjustment was performed for multiple comparison purposes. Significance level was set at $p \leq 0.05$.

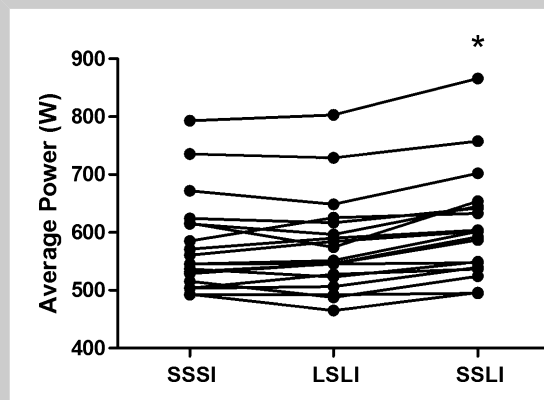


Figure 1. Individual average power (W) values across the experimental conditions. * $p < 0.05$ when compared with both SSSI and LSLI.

TABLE 2. Average muscle power and blood lactate concentration during the experimental conditions.*

	SSSI	LSLI	SSLI
Power (Watts)	581.86 ± 113.18	578.13 ± 138.78	601.88 ± 142.48 [†]
Lac pre (mmol·l ⁻¹)	0.59 ± 0.27	0.56 ± 0.22	0.53 ± 0.2
Lac post (mmol·l ⁻¹)	1.77 ± 0.86 [‡]	2.28 ± 0.96 [‡]	1.05 ± 0.49 [§]

*SSSI = short-set short-interval; LSLI = long-set long-interval; SSLI = short-set long-interval; Lac = blood lactate concentration.

[†] $p < 0.05$ when compared with SSSI and LSLI.

[‡] $p < 0.05$ when compared with LSLI.

[§] $p < 0.05$ when compared with Pre values (within-group comparisons).

RESULTS

Average power was significantly greater in the SSLI (lower exercise to rest ratio group) when compared with the exercise to rest ratio-equated conditions (i.e., SSSI and LSLI) (Figure 1).

Blood lactate concentration was significantly increased over time (pre vs. post) only in the exercise to rest ratio-equated conditions (i.e., SSSI and LSLI). Further, the SSLI condition (lower exercise to rest ratio) resulted in significantly lower blood lactate concentration when compared with LSLI. Average power and blood lactate concentration data are presented in Table 2.

DISCUSSION

The main finding of the present study is that the lower exercise to rest ratio protocol (SSLI) resulted in greater average power production in resistance exercise sessions aiming at developing muscle power production capacity. Additionally, both the exercise to rest ratio-equated conditions presented similar performance and metabolic results. Thus, shorter rest intervals may effectively maintain average power production if a fewer number of repetitions per set is used (SSSI) in a similar fashion than longer sets with longer intervals (LSLI).

The exercise to rest ratio may be manipulated by changing the exercise volume within a set (number of repetitions per set) or the rest interval between sets or repetitions. In this concern, previous studies have shown that for a given effort, shorter rather than longer rest intervals (i.e., higher exercise to rest ratio protocols) may hamper the maintenance of strength-related parameters (e.g., maximum number of repetitions, peak torque, and work) (1,14,16–18).

Regarding muscle power output, our findings are in agreement with those by Abdessemed et al. (1). These authors evaluated 3 different rest interval schemes in nonequated exercise to rest ratio conditions (using the same number of repetitions per set between conditions; 10 sets of 6 repetitions with 70% 1RM). Higher exercise to rest ratio conditions (i.e., shorter rest interval for a given exercise

volume) resulted in a significant decrease in power throughout multiple exercise sets. In our study, the comparison between the nonequated conditions (i.e., SSLI vs. SSSI and SSLI vs. LSLI) led to a similar conclusion, with the SSLI (lower exercise to rest ratio condition) presenting better results even when the resistance exercise was muscle power development oriented. It is important to note that the higher exercise to rest ratio condition performed better when either the

rest interval was manipulated (i.e., SSLI vs. SSSI) or when the number of repetitions was increased while maintaining a constant rest interval between conditions (i.e., SSLI vs. LSLI). Pincivero et al. (12) further support these findings. In nonequated exercise to rest ratio conditions, the shorter the rest interval (for the same given effort; i.e., 4 sets of 20 maximal isokinetic concentric knee extensions), the greater the impairment of power production. Despite the evident differences between the exercise protocols between ours and the aforementioned studies (i.e., exercise contraction mode and intensity), it seems reasonable to suggest that the exercise to rest ratio plays an important role in power maintenance.

However, when exercise to rest ratio is equated, the exercise loading scheme is thought to affect muscle power output. Lawton et al. (9) compared 3 different exercise to rest ratio-equated conditions with different schemes of rest interval and number of repetitions per set (2 sets of 3 repetitions with 100-second intervals, 3 sets of 2 repetitions with 50-second intervals, and 6 sets of 1 repetition with 20-second intervals) with a traditional (nonequated) high-intensity continuous-resistance exercise protocol of equal volume (i.e., total of 6 repetitions) and intensity (i.e., load correspondent to 6 repetition maximum [6RM]) but different exercise to rest ratio (i.e., 1 single set of 6RM). Similarly to our findings, the authors observed that a shorter number of repetitions per set requires a shorter rest interval for muscle power maintenance. Accordingly, Denton and Cronin (7) observed that reducing the rest interval in a shorter-set loading scheme (fewer repetitions per set) results in similar power output when compared with a exercise to rest ratio-equated condition of longer sets and intervals (i.e., 8 sets of 3 repetitions with a 130-second interval vs. 4 sets of 6 repetitions with a 302-second interval). Importantly, both studies used a relatively greater exercise load (i.e., 6RM) when compared with ours (i.e., 60% 1RM). Traditionally, muscle power training usually comprises loads between 30 and 60% 1RM. Therefore, it seems reasonable to suggest that our findings further extend this notion to more specific muscle power training conditions.

Blood lactate concentration data revealed that the exercise to rest ratio differently affects metabolic responses to exercise. The SSLI condition resulted in significantly lower postexercise blood lactate concentration when compared with both the SSSI and the LLLI conditions. These data suggest that the higher average power observed during SSLI may be related to a lesser reliance on glycolysis (e.g., given the lower blood lactate concentration), because muscle phosphorylcreatine content may have been more effectively replenished during the rest intervals (2,11).

Despite the interesting findings herein, caution should be exercised when interpreting and extrapolating these data. It is not known if different loads, different type of lifts, or different training programs and loading schemes will produce similar results as those observed in the present study.

In summary, these findings suggest that in the present experimental conditions, shorter rest intervals may fully restore the individual's ability to produce muscle power if a smaller exercise volume per set is performed and that lower exercise to rest ratio protocols result in greater average power production when compared with higher ratio ones.

PRACTICAL APPLICATIONS

Strength training coaches are often concerned about the specificity of their loading schemes in relation to their sport. In this regard, the manipulation of the exercise volume and the rest interval between exercise sets plays an important role in a task-specific training design. The present study demonstrated that when using the squat exercise and similar loads and loading schemes to those in the present study, coaches can successfully reduce their rest intervals to attend a task-specific demand and still maintain muscle power output if a concomitant reduction in exercise volume is performed. Thus, coaches should consider the exercise to rest ratio to increase the volume and the intensity of power-oriented training loads throughout a training program. Increasing the training intensity by decreasing the duration of the rest interval between sets may hamper power production. Similarly, augmenting the training volume, while maintaining the exercise to rest ratio, may diminish the power production in the last training sets. It is possible to suggest that larger increments in training volume should be followed by a lower exercise to rest ratio to allow a complete recover between sets. Furthermore, a low exercise to rest ratio should always be used to maximize power production between the sets of power exercises, as a high exercise to rest ratio seems to decrease muscle power production capacity.

ACKNOWLEDGMENTS

Hamilton Roschel is supported by Fundação de Amparo a Pesquisa do Estado de São Paulo (FAPESP; 2010/51428-2). Carlos Ugrinowitsch is supported by Conselho Nacional de

Desenvolvimento Científico e Tecnológico (CNPq; 470207/2008-6 and 303162/2008-2).

REFERENCES

1. Abdessemed, D, Duche, P, Hautier, C, Poumarat, G, and Bedu, M. Effect of recovery duration on muscular power and blood lactate during the bench press exercise. *Int J Sports Med* 20: 368–373, 1999.
2. Balsom, PD, Soderlund, K, Sjodin, B, and Ekblom, B. Skeletal muscle metabolism during short duration high-intensity exercise: Influence of creatine supplementation. *Acta Physiol Scand* 154: 303–310, 1995.
3. Bosco, C, Belli, A, Astrua, M, Tihanyi, J, Pozzo, P, Kellis, S, Tsarpela, O, Foti, C, Manno, R, and Tranquilli, C. A dynamometer for evaluation of dynamic muscle work. *Eur J Appl Physiol Occup Physiol* 70: 379–386, 1995.
4. Brown, LE and Weir JP. ASEP procedures recommendation I: Accurate assessment of muscular strength and power. *J Exerc Physiol* 4: 1–21, 2001.
5. Cronin, JB, McNair, PJ, and Marshall, RN. The role of maximal strength and load on initial power production. *Med Sci Sports Exerc* 32: 1763–1769, 2000.
6. De Salles, BF, Simao, R, Miranda, F, Novaes Jda, S, Lemos, A, and Willardson, JM. Rest interval between sets in strength training. *Sports Med* 39: 765–777, 2009.
7. Denton, J and Cronin, JB. Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res* 20: 528–534, 2006.
8. Hill-Haas, S, Bishop, D, Dawson, B, Goodman, C, and Edge, J. Effects of rest interval during high-repetition resistance training on strength, aerobic fitness, and repeated-sprint ability. *J Sports Sci* 25: 619–628, 2007.
9. Lawton, TW, Cronin, JB, and Lindsell, RP. Effect of interrepetition rest intervals on weight training repetition power output. *J Strength Cond Res* 20: 172–176, 2006.
10. Legaz-Arrese, A, Reverter-Masia, J, Munguia-Izquierdo, D, and Ceballos-Gurrola, O. An analysis of resistance training based on the maintenance of mechanical power. *J Sports Med Phys Fitness* 47: 427–436, 2007.
11. McMahon, S, and Jenkins, D. Factors affecting the rate of phosphocreatine resynthesis following intensive exercise. *Sports Med* 32: 761–784, 2002.
12. Pincivero, DM, Gear, WS, Moyna, NM, and Robertson, RJ. The effects of rest interval on quadriceps torque and perceived exertion in healthy males. *J Sports Med Phys Fitness* 39: 294–299, 1999.
13. Ratamess, NA, Alvar, BA, Evetoch, TK, Housh, TJ, Kibler, WB, and Kraemer, WJ. American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc* 41: 687–708, 2009.
14. Robinson, JM, Stone, MH, Johnson, RL, Penland, CM, Warren, BJ, and Lewis, RD. Effects of different weight training exercise/rest intervals on strength, power, and high intensity exercise endurance. *J Strength Cond Res* 9: 216–221, 1995.
15. Willardson, JM. A brief review: factors affecting the length of the rest interval between resistance exercise sets. *J Strength Cond Res* 20: 978–984, 2006.
16. Willardson, JM and Burkett, LN. A comparison of 3 different rest intervals on the exercise volume completed during a workout. *J Strength Cond Res* 19: 23–26, 2005.
17. Willardson, JM and Burkett, LN. The effect of rest interval length on bench press performance with heavy vs. light loads. *J Strength Cond Res* 20: 396–399, 2006.
18. Willardson, JM and Burkett, LN. The effect of different rest intervals between sets on volume components and strength gains. *J Strength Cond Res* 22: 146–152, 2008.